

```
!pip install nltk
import nltk
from nltk.tokenize import TweetTokenizer
from nltk.corpus import twitter_samples
```

Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
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 Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.

```
nltk.download('twitter_samples')
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package twitter_samples to /root/nltk_data...
[nltk_data]   Unzipping corpora/twitter_samples.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
True
```

```
tweets=twitter_samples.strings('positive_tweets.json')
for i in range(3):
    print("Tweet",i+1)
    print(tweets[i])
    print()
```

Tweet 1
 #FollowFriday @France_Inte @PKuchly57 @Milipol_Paris for being top engaged members in my communit

Tweet 2
 @Lamb2ja Hey James! How odd :/ Please call our Contact Centre on 02392441234 and we will be able

Tweet 3
 @DespiteOfficial we had a listen last night :) As You Bleed is an amazing track. When are you in

tokenization using tweettokenizer

```
tokenizer=TweetTokenizer(
    preserve_case=False,
    strip_handles=True,
    reduce_len=True
)
tokenized_tweets=[tokenizer.tokenize(tweet) for tweet in tweets[:5]]
for i,tokens in enumerate(tokenized_tweets):
    print("Tweet",i+1,"Tokens:")
    print(tokens)
    print()
```

Tokens:
 wfriday', 'for', 'being', 'top', 'engaged', 'members', 'in', 'my', 'community', 'this', 'week', '

Tokens:
 'james', '!', 'how', 'odd', ':/', 'please', 'call', 'our', 'contact', 'centre', 'on', '02392441234

Tokens:
 had', 'a', 'listen', 'last', 'night', ':)', 'as', 'you', 'bleed', 'is', 'an', 'amazing', 'track',

Tokens:
 its', ':)']

Tokens:

!, 'yipppy', '!', '!', '!', 'my', 'acctnt', 'verified', 'rqst', 'has', 'succeed', 'got', 'a', 'blu

pos tagging using nltk

```
pos_tagged_tweets=[nltk.pos_tag(tokens) for tokens in tokenized_tweets]
for i,tagged in enumerate(pos_tagged_tweets):
    print("Tweet",i+1,"POS Tags:")
    print(tagged)
    print()
```

Tweet 1 POS Tags:

[('#followfriday', 'NN'), ('for', 'IN'), ('being', 'VBG'), ('top', 'JJ'), ('engaged', 'VBN'), ('m

Tweet 2 POS Tags:

[('hey', 'NN'), ('james', 'NNS'), ('!', '.'), ('how', 'WRB'), ('odd', 'JJ'), (':/', 'JJ'), ('plea

Tweet 3 POS Tags:

[('we', 'PRP'), ('had', 'VBD'), ('a', 'DT'), ('listen', 'VBN'), ('last', 'JJ'), ('night', 'NN'),

Tweet 4 POS Tags:

[('congrats', 'NNS'), (':)', 'VBP')]

Tweet 5 POS Tags:

[('yeeaaah', 'NN'), ('yipppy', 'JJ'), ('!', '.'), ('!', '.'), ('!', '.'), ('my', 'PRP\$'), ('acctnt'

pos tagging on custom noisy text

```
text="omg I luv this phone 🥰 #awesome #AI"
tokens=tokenizer.tokenize(text)
tags=nltk.pos_tag(tokens)
print("original Text:",text)
print("Tokens:",tokens)
print("POS Tags:",tags)
```

original Text: omg I luv this phone 🥰 #awesome #AI

Tokens: ['omg', 'i', 'luv', 'this', 'phone', '🥰', '#awesome', '#ai']

POS Tags: [('omg', 'NN'), ('i', 'NN'), ('luv', 'VBP'), ('this', 'DT'), ('phone', 'NN'), ('🥰', 'V

extract nouns and verbs

```
nouns=[]
verbs=[]
for word,tag in tags:
    if tag.startswith('NN'):
        nouns.append(word)
    elif tag.startswith('VB'):
        verbs.append(word)
print("Nouns:",nouns)
print("Verbs:",verbs)
```

Nouns: ['omg', 'i', 'phone', '#awesome', '#ai']

Verbs: ['luv', '🥰']

